

AI LEGAL™

Student Learning Portal & Gamification Dashboard

Comprehensive Product, Technical & Architecture Documentation

Document Area	Details
Primary Component	StudentDashboardSection
Route	/dashboard
Primary Experience	Learning Dashboard → Challenges → Study Tools → Analytics
Core Concept	Personalized Academic Headquarters & Gamified Study Management
Target Users	Law Students, Judicial Aspirants & Bar Examinees

Document Structure

- 1. Executive Summary & Module Purpose
- 2. Core Value Propositions
- 3. Dashboard Architectural Layout
- 4. Deep Feature Breakdown
- 5. Technical Implementation Details
- 6. Cross-Module Centrality
- 7. Functional Requirements & UX Behaviour
- 8. Study Data & State Model
- 9. Future Extensions

1. Executive Summary & Module Purpose

The Student Learning Portal & Gamification Dashboard, rendered through StudentDashboardSection within /dashboard, serves as the personalized academic headquarters for law students, judicial aspirants, and bar examinees using AI LEGAL™.

Where an Advocate dashboard is centered around litigation workflows, dockets, hearings, and case management, the Student Learning Portal is organized around daily learning consistency, exam readiness, curriculum progression, study analytics, and habit-forming gamification.

The dashboard acts as a central routing layer connecting the student's primary learning tools: AI Knowledge Hub, Quiz & MCQ Practice, AI Notes Maker, Legal Precedents & Research Assistant, and AI Legal Chat Assistant.

2. Core Value Propositions

Value Proposition	Description
Study Streak (🔥)	Tracks consecutive active study days and provides motivational states based on continued activity.
Composite Study Progress	Combines completed learning tools and saved matters into a visible progress percentage.
Today's Challenge	Provides three daily high-yield learning activities spanning quizzes, comparative bare-act study, and landmark judgment analysis.
Continue Learning	Tracks recent Knowledge Hub reading sessions and enables one-click study resumption.
Academic Quota & Analytics	Provides transparent visibility into tool usage, active matters, precedent research usage, and estimated study hours.

3. Dashboard Architectural Layout

Dashboard Area	Primary Elements
1. Today's Learning Header	Welcome/student identifier, Product Guide launcher, Knowledge Hub Bookshelf action.
2. Study Progress Card	Progress percentage, completed tools, saved matters.
3. Study Streak Card	Current streak, daily goal status, momentum banner.
4. Continue Learning	Recent statute, reading progress, Resume Study action.
5. Quick Learning Actions	AI Notes Workspace and Quiz & MCQ Practice.
6. Today's Challenges	Daily Quiz, Bare Act Comparative Breakdown, Landmark Judgment Ratio.
7. Learning Analytics & Quota	Tool usage, active matters, precedent research, estimated study hours.

3.1 Dashboard Navigation Model

- The header provides an immediate orientation point and direct access to the Knowledge Hub Bookshelf.
- Progress and streak cards expose daily learning status without requiring navigation into individual tools.
- Continue Learning reduces friction by restoring the student's latest statutory reading context.
- Quick actions provide direct entry points into high-frequency study tools.
- Today's Challenges turn the dashboard into a daily task hub rather than a passive analytics screen.

- Analytics and quota monitors provide visibility into usage and learning activity.

4. Deep Feature Breakdown

4.1 Study Streak Engine — `streakService.js`

- A study activity event can be triggered when a student accesses the Knowledge Hub, generates a note, attempts a quiz, or conducts precedent research.
- `recordStudyActivity(userId)` records the activity timestamp and updates the user's streak state.
- Yesterday activity: the active streak increments by one.
- Already studied today: the streak remains active without double-incrementing.
- Extended inactivity: after the configured inactivity threshold, the streak resets to 1.
- Motivational states can be associated with streak length, for example: 1 Day — “Great start today!”, 2–6 Days — “On fire!”, and 7+ Days — “Unstoppable momentum!”

4.2 Composite Study Progress Metric

The dashboard combines completed learning tools and saved matters into a normalized progress percentage:

$$\text{Progress \%} = \min(100, ((\text{Completed Tools} + \text{Saved Matters}) / (\text{Tools Limit} + \text{Matters Limit})) \times 100)$$

- The documented example uses a Tools Limit of 5 and a Matters Limit of 3.
- The metric is capped at 100% to prevent over-completion from producing a value above the maximum.
- The resulting percentage is surfaced as a visual progress target for the student's learning activity.

4.3 Today's Challenge Engine

Challenge	Purpose	Destination
Daily Quiz Challenge	5 high-yield Constitutional Law / Fundamental Rights MCQs.	/dashboard/tools/quiz-practice
Bare Act Comparative Challenge	Historic vs. newly enacted law comparison, such as IPC ↔ BNS.	/dashboard/tools/knowledge-hub
Landmark Precedent Challenge	Review a foundational Supreme Court precedent and its legal ratio.	/dashboard/tools/legal-precedents

4.4 Learning Analytics & Subscription Quota Tracker

- Tool Usage Progress: Real-time percentage of student learning features utilized.
- Active Matters: Tracks saved student mock dossiers against the applicable plan limit.
- Precedent Research: Tracks case-law queries and statutory lookups against student plan limits.
- Estimated Study Hours: Dynamically estimates learning time from session interactions and tool-completion weights.

5. Technical Implementation Details

Layer	Implementation	Purpose
Web Frontend	<code>StudentDashboardSection.jsx</code>	Primary student dashboard component.
Mobile Frontend	<code>dashboard/index.tsx</code>	Mobile dashboard when the student role is selected.

Layer	Implementation	Purpose
Streak Service	streakService.js	Activity timestamps, streak counters, reset logic, and persistence.
Role State	userData.js	Recoil selector providing selectedRoleState === 'student'.
Product Guide	StudentProductGuideModal.jsx	Interactive walkthrough of student capabilities.
Persistence	localStorage + backend user metadata	Stores activity and relevant student dashboard state.

6. Cross-Module Centrality

The Student Learning Portal operates as the routing orchestrator for the student ecosystem. It does not replace the specialized learning tools; instead, it connects them into a single daily learning journey.

- AI Knowledge Hub: Opens bare-act and statutory reading workflows.
- Quiz & MCQ Practice: Routes students to assessment and exam simulation.
- AI Notes Maker: Routes students to synthesis, revision notes, and case briefs.
- Legal Precedents & Research Assistant: Opens judgment and case-law research.
- AI Legal Chat Assistant: Can open with prefilled study-topic prompts for interactive doubt resolution.

7. Functional Requirements & UX Behaviour

Requirement	Expected Behaviour
Role-Aware Dashboard	Student-specific dashboard should render when selectedRoleState is set to student.
Streak Integrity	Repeated activity on the same day should not incorrectly increment the streak more than once.
Progress Calculation	Progress should be normalized and capped at 100%.
Challenge Rotation	Daily challenges should be presented as discrete actionable tasks with direct deep links.
Resume Learning	Recent Knowledge Hub activity should be available for one-click continuation.
Quota Transparency	Usage counters and limits should be clearly visible to the student.
Product Guidance	StudentProductGuideModal should provide an accessible walkthrough of major capabilities.
Responsive UX	Dashboard cards, challenge components, and analytics should remain usable across web and mobile layouts.

8. Study Data & State Model

The student dashboard can be represented through the following conceptual state flow:

User Role → Study Activity → Streak State → Tool Completion → Saved Matters → Progress Metric → Challenges → Analytics / Quota → Learning Tool

Activity events originate from learning tools and are aggregated by the dashboard. The resulting state is then used to render streaks, progress, challenges, quota indicators, and continuation actions.

9. Future Extensions

- Weekly and monthly study reports with subject-level mastery trends.
- Adaptive Today's Challenges based on recent quiz errors and Knowledge Hub reading history.
- Personalized study-plan generation based on examination date and target syllabus.
- Calendar-based study scheduling and deadline reminders.
- Achievement badges for milestones such as completed quizzes, saved notes, and consecutive study days.
- Subject mastery heatmaps across Constitutional, Criminal, Civil, Evidence, and Commercial Law.
- Cross-device synchronization of streaks, progress, bookmarks, and recent learning sessions.

Appendix A — Example Daily Student Journey

1. Student opens the dashboard and sees current progress and streak.
2. Student resumes the most recent Knowledge Hub section.
3. Student completes the Daily Quiz Challenge.
4. Student reviews the Bare Act Comparative Challenge.
5. Student studies the Landmark Judgment Challenge.
6. Activity is recorded through the study activity service.
7. Dashboard updates streak, progress, usage analytics, and estimated study hours.
8. Student moves into Notes Maker or Legal Precedents for deeper study.

Appendix B — Reference Implementation Paths

Area	Reference
Web Dashboard	AI-Legal_App_Webapp/src/Components/StudentDashboardSection.jsx
Mobile Dashboard	AI-Legal_App_frontend-mobile/src/app/(tabs)/dashboard/index.tsx
Streak Service	streakService.js
Role State	userData.js
Product Guide	AI-Legal_App_Webapp/src/Components/StudentProductGuideModal.jsx